

Thermodynamics of $T\bar{T}$ -deformed $O(N)$ vector model

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2021/1/25 @ KAWS-2021

Based on work in progress with K. Sakai, K. Yoshida
c.f. arXiv:2002.01414 with T. Ishii, H. Kawai, K. Sakai, K. Yoshida

What is $T\bar{T}$ -deformation?

- An irrelevant deformation of 2D QFT

[F.Smirnov and A.Zamolodchikov | 608.05499]

[A.Cavagli, S.Negro, I.Szcsnyi and R.Tateo | 608.05534]

- It follows “ $T\bar{T}$ flow equation”

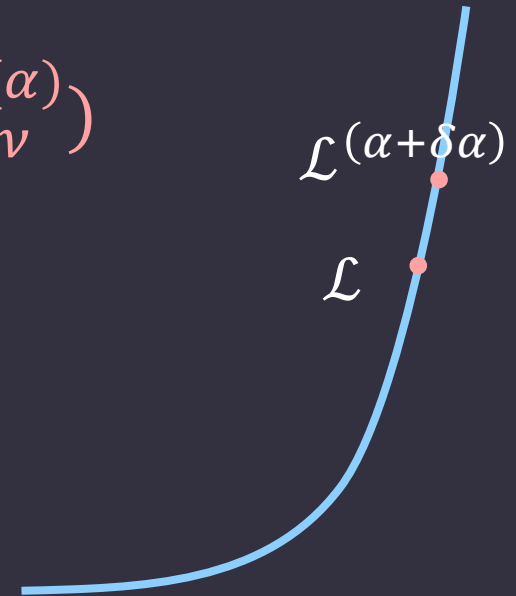
$$\mathcal{L}^{(\alpha+\delta\alpha)} = \mathcal{L}^{(\alpha)} - \delta\alpha \det(T_{\mu\nu}^{(\alpha)})$$

$$\Leftrightarrow \frac{d}{d\alpha} \mathcal{L}^{(\alpha)} = -\det(T_{\mu\nu}^{(\alpha)})$$

$T_{\mu\nu}^{(\alpha)}$: Energy-Momentum Tensor of $\mathcal{L}^{(\alpha)}$

α : deformation parameter

$$[\alpha] = M^{-2}$$



Features

- $T\bar{T}$ -deformation of free massless $O(N)$ vector model = Nambu-Goto action in static gauge

[G.Bonelli, N.Doroud, M.Zhu | 804.10967]

[A.Cavaglià, S.Negro, I.Szécsényi, R.Tateo | 608.05534]

$$\mathcal{L}^{(\alpha)} = \sqrt{-\det h_{\mu\nu}}, \quad h_{\mu\nu} = \frac{\eta_{\mu\nu}}{\sqrt{2\alpha}} + \partial_\mu \vec{\phi} \cdot \partial_\nu \vec{\phi}$$

- $T\bar{T}$ -deformed Energy spectrum on $R^1 \times S^1$ satisfies the forced inviscid Burgers equation

$$\frac{\partial}{\partial \alpha} E_n^{(\alpha)} = E_n^{(\alpha)} \frac{\partial}{\partial R} E_n^{(\alpha)} - \frac{P_n^2}{R^2} \quad x \sim x + R$$

P_n ...momentum of the eigenstate

- And more.. (non-singularity of $\det T_{\mu\nu}$, factorization, deformed S-matrix, Gravity dual, etc)

Motivation and Our Work

- $T\bar{T}$ -deformation has many interesting features
- But we know that irrelevant theories should have some inconsistencies as QFT

(c.f. [JH, T. Ishii, H. Kawai, K. Sakai, K. Yoshida 2002.01414])

- We want to know why $T\bar{T}$ -deformation works well
- We calculate the thermal free energy of the $T\bar{T}$ -deformed free massless $O(N)$ vector model in two ways:
the path integral and the Burgers equation

Results

$\frac{\alpha}{N}$... deformation parameter
 β ... inversed temperature

- By Path integral, we get the **leading contribution** in the large N limit:

$$f_{PI} = \frac{N}{2\alpha} \left(-1 + \sqrt{1 - \frac{2\pi^2}{3} \frac{\alpha}{\beta^2}} \right)$$

- It can be obtained **exactly** from $T\bar{T}$ -deformed ground state energy with the Burgers eq.:

$$f_{Burg} = \frac{N}{2\alpha} \left(-1 + \sqrt{1 - \frac{2\pi^2}{3} \frac{\alpha}{\beta^2}} \right)$$

Summary and Discussion

- Results are **the same**
- But the result from the **large N limit** has **$1/N$ corrections**, while one from the Burgers eq. **does not**
- This fact suggests that there are **miraculous cancellations** of $1/N$ corrections
- We are going to investigate this point further

Thank you!

Summary of today's talk

- We investigate an irrelevant deformation of 2D QFT, called " $T\bar{T}$ -deformation"

In particular, $T\bar{T}$ -deformation of the free massless $O(N)$ vector model is the Nambu-Goto action in the static gauge

- We computed the thermal free energy from the path integral in the large N limit and the Burgers equation

Example of $T\bar{T}$ deformation

- We give a examples of $T\bar{T}$ -deformed theories
- We begin with the free massless scalar

$$\mathcal{L}^{(0)} = -(\partial_\mu \phi)^2$$

- $T\bar{T}$ deformation of this model is

$$\mathcal{L}^{(\alpha)} = \frac{1}{2\alpha} \left(-1 + \sqrt{1 - 4\alpha (\partial_\mu \phi)^2} \right)$$

Example of $T\bar{T}$ deformation

- Consider the free massless $O(N)$ vector model

$$\mathcal{L}^{(0)} = -(\partial_\mu \vec{\phi})^2 \quad \left[\begin{array}{l} \phi_i \dots \text{real scalars} \\ i = 1, \dots, N \end{array} \right]$$

- $T\bar{T}$ -deformation of this model is

$$\mathcal{L}^{(\alpha)} = \frac{1}{2\alpha} \left(-1 + \sqrt{1 - 4\alpha(\partial_\mu \vec{\phi})^2 - 16\alpha^2 \left((\partial_0 \vec{\phi})^2 (\partial_1 \vec{\phi})^2 - (\partial_0 \vec{\phi} \cdot \partial_1 \vec{\phi})^2 \right)} \right)$$

[G.Bonelli, N.Doroud, M.Zhu | 804.10967]

[A.Cavaglià, S.Negro, I.Szécsényi, R.Tateo | 608.05534]

Nambu-Goto in static gauge

- Surprisingly, the resulting action is the **Nambu-Goto action**

$$\mathcal{L}_{NG} = \sqrt{-\det h_{\mu\nu}}$$

in the static gauge

$$\left(\begin{array}{l} h_{\mu\nu} := \gamma_{ij} \partial_\mu \phi^i \partial_\nu \phi^j \\ i, j = \tau, \sigma, 1, \dots, N \\ \gamma_{ij} := \text{diag}(-, +, \dots, +) \end{array} \right)$$

$$\phi^\tau = \frac{x^0}{\sqrt{2\alpha}}, \phi^\sigma = \frac{x^1}{\sqrt{2\alpha}}$$

$\Rightarrow$ related to **string theory at classical level**

(Note: the dimensions of target space are **N+2**)

Good properties of $\det T_{\mu\nu}$

[A.B.Zamolodchikov hep-th/04011146]

- $\det T_{\mu\nu}$ is a **composite** operator, but can be **defined well** as a **local** operator

$$\det T_{\mu\nu}(z, w) = \hat{O}_{T\bar{T}}(z) + (\text{total derivative})$$

- Its expectation value in energy eigenstates **factorizes** into products of **one point functions**

$$\langle \det T_{\mu\nu} \rangle_n = \langle T_{00} \rangle_n \langle T_{11} \rangle_n - \langle T_{01} \rangle_n \langle T_{10} \rangle_n$$

$$\langle \mathcal{O} \rangle_n := \langle n | \mathcal{O} | n \rangle$$

$T\bar{T}$ -deformed energy spectrum

- Hamiltonian of $T\bar{T}$ -deformed theories satisfies

$$\frac{d}{d\alpha} H^{(\alpha)} = \det T_{\mu\nu}^{(\alpha)}$$

- If the spacetime is a infinitely long cylinder, we can calculate the deformed energy spectrum from the Burgers equation

[A.B.Zamolodchikov 04011146]

$$\frac{\partial}{\partial \alpha} E_n^{(\alpha)} = E_n^{(\alpha)} \frac{\partial}{\partial R} E_n^{(\alpha)} - \frac{P_n^2}{R^2}$$

$$x \sim x + R$$

P_n ...momentum of the eigenstate

(Note: By multiplying $\exp(-\beta E_n^{(\alpha)})$ and taking sum about all energy eigenstates, we get the $T\bar{T}$ flow equation)

Background

- $T\bar{T}$ -deformation has many interesting features
- But we know that irrelevant theories should have some inconsistencies as QFT
- For example, we showed that $T\bar{T}$ -deformed free $O(N)$ vector model in the large N limit has a negative-norm state

[JH, T. Ishii, H. Kawai, K. Sakai, K. Yoshida 2002.01414]

Motivation

- In the previous paper, we studied the model on the flat spacetime
- But we cannot compare the result with one from the Burgers equation
- We compute some physical quantities on a cylinder

Our work

- We study thermodynamics of the $T\bar{T}$ -deformed free massless $O(N)$ vector model
- The thermal free energy can be calculated in two ways: the path integral and the Burgers equation

Path integral

- Here we calculate the thermal free energy by the path integral
- From the point of view of a perturbation theory, the Lagrangian has infinitely many interactions

$$\mathcal{L} = -(\partial_\mu \vec{\phi})^2 - \alpha \left[(\partial_\mu \vec{\phi})^4 - 4 \left((\partial_0 \vec{\phi})^2 (\partial_1 \vec{\phi})^2 - (\partial_0 \vec{\phi} \cdot \partial_1 \vec{\phi})^2 \right) \right] + O(\alpha^2)$$

- We consider the large N limit

$$N \rightarrow \infty, \quad \alpha N: \text{fixed}$$

Auxiliary fields

- With auxiliary fields $C^{\mu\nu}$, the Lagrangian is rewritten in a quadratic form of $\vec{\phi}$

$$\mathcal{L} = \frac{N}{4\alpha} \sqrt{-C} C^{\mu\nu} h_{\mu\nu} - \frac{N}{2\alpha}$$

$C^{\mu\nu}$... a symmetric auxiliary field

(Note: We rescaled $\alpha \rightarrow \alpha/N$)

- After integrating $\vec{\phi}$ out, we get

$$S_{eff} = \frac{iN}{2} \text{tr} \log(-\partial_\mu \sqrt{-C} C^{\mu\nu} \partial_\nu) + \frac{N}{2\alpha} \int dx \left(\frac{1}{2} \sqrt{-C} C^{\mu\nu} \eta_{\mu\nu} - 1 \right)$$

as the effective action for $C^{\mu\nu}$

Free energy in large N

- In the large N limit, integration of $C^{\mu\nu}$ is equivalent to substituting the values at the stationary point of S_{eff}
- The thermal free energy density f_{PI} is related to S_{eff} as

$$f_{PI} = \frac{1}{i \int dx} S_{eff} \Big|_{stationary\ point}$$

- After calculation, we get

$$f_{PI} = \frac{N}{2\alpha} \left(-1 + \sqrt{1 - \frac{2\pi^2}{3} \frac{\alpha}{\beta^2}} \right)$$

Calculation with Burgers eq.

- Here we calculate the thermal free energy with the Burgers equation
- The crucial point is the Zamolodchikov's argument in his old paper

[A.B. Zamolodchikov Nucl. Phys. B 342 (1990) 695–720]

- It showed that the thermal free energy of a theory on $R^{1,1}$ with the inverse temperature β is related to the ground state energy of the theory on a cylinder with a circumference β

Zamolodchikov's argument

- Thermal free energy can be calculate from

$$Z = \int D\psi \exp \left(- \int_{R^1 \times S^1} d^2 x_E S_E \right) = \exp \left(- \int d^2 x_E f^{(R^1,1)} \right)$$

- If we exchange roles of the (Euclidean) time and space, we get

$$Z = \exp \left(- \int dx_1 E_0^{(R^1 \times S^1)} \right) = \exp \left(- \int d^2 x_E \frac{E_0^{(R^1 \times S^1)}}{R} \right)$$

- Then we conclude

$$f^{(R^1,1)} = E_0^{(R^1 \times S^1)} / R \Big|_{R=\beta}$$

Free energy from Burgers eq.

- With the result of N free massless scalars on a cylinder, we get the ground state energy from the Burgers eq.

$$\left. \frac{E_0}{R} \right|_{\alpha=0} = -\frac{\pi^2}{6} \frac{N}{R^2} \quad \longrightarrow \quad \frac{E_0}{R}(\alpha/N) = \frac{N}{2\alpha} \left(-1 + \sqrt{1 - \frac{2\pi^2}{3} \frac{\alpha}{R^2}} \right)$$

- With the Zamolodchikov's argument, we get

$$f_{Burg} = \frac{N}{2\alpha} \left(-1 + \sqrt{1 - \frac{2\pi^2}{3} \frac{\alpha}{\beta^2}} \right)$$

Summary

- Studied an irrelevant deformation of 2D quantum field theories, called " $T\bar{T}$ - deformation"
- Computed the thermal free energy from the path integral and the Burgers equation
- The results are identical at leading order in the large N limit

Thank you!

- 正準形式での経路積分

$$Z_{can} = \int D\vec{\phi} D\vec{\pi} \exp \left(i \int dx \tilde{\mathcal{L}} \right)$$

$$\tilde{\mathcal{L}} := \vec{\pi} \cdot \partial_0 \vec{\phi} - \frac{1}{2\alpha} \left(1 - \sqrt{1 - 4\alpha \vec{\pi}^2 - 4\alpha (\partial_1 \vec{\phi})^2 + 16\alpha^2 (\vec{\pi} \cdot \partial_1 \vec{\phi})^2} \right)$$

- 補助場の導入

$$\tilde{\mathcal{L}} = \frac{N}{2\alpha} \left(\vec{\pi} \cdot \partial_0 \vec{\phi} + \lambda \left(\pi^2 + (\partial_1 \vec{\phi})^2 - 2\sigma \vec{\pi} \cdot \partial_1 \vec{\phi} \right) + \lambda(\sigma^2 - \rho) - 1 + \sqrt{1 - \rho} \right) - \frac{i}{2} \text{tr} \log \lambda$$

- 有効作用を書き下す

$$Z_{can} = \int D\vec{\phi} D\vec{\pi} D\lambda D\rho D\sigma \exp\left(i \int dx \tilde{\mathcal{L}}\right)$$

$$= \int D\lambda D\rho D\sigma \exp(iS_{eff})$$

$$S_{eff} := \frac{iN}{2} \text{tr} \log \left(-\partial_0 \lambda \partial_0 + \frac{1}{4} (\partial_0 - \lambda \sigma \partial_1)^2 \right) + \frac{N}{2\alpha} \int dx \left(\lambda (\sigma^2 - \rho) - 1 + \sqrt{1 - \rho} \right)$$

- 停留点を代入して自由エネルギーの表式

$$f_{can} = \frac{1}{2\alpha} \left(-1 + \sqrt{1 - \frac{2\pi^2}{3} \frac{\alpha}{\beta^2}} \right)$$

How to resolve?

- Possible candidates

1. The canonical formalism of the path integral differs from the Lagrangian formalism?
2. $1/N$ corrections cancel out at all-order?
3. The Burgers equation or the operator formalism cannot capture $1/N$ corrections?

Hamiltonian and Path integral

- Hamiltonian of static NG model is

$$\mathcal{H}_{NG} = \frac{1}{2\alpha} \left(1 - \sqrt{1 - 4\alpha(\partial_1 \vec{\phi})^2 - \alpha \vec{\pi}^2 + 4\alpha^2(\vec{\pi} \cdot \partial_1 \vec{\phi})^2} \right)$$

- Hamiltonian is NOT a quadratic form of $\vec{\pi}$

⇒ The canonical and Lagrangian formalism of the path integral are NOT equal generally

$$\int D\vec{\phi} D\vec{\pi} \exp \left(i \int dx (\vec{\pi} \cdot \partial_0 \vec{\phi} - \mathcal{H}) \right) \neq \int D\vec{\phi} \exp \left(i \int dx \mathcal{L} \right)$$

Free energy in canonical PI

- With the large N method, we can calculate the free energy in the canonical formalism of the path integral as well as in the Lagrangian formalism
- But **the result coincides** with the one in the Lagrangian formalism

$$f_{can} = \frac{N}{2\alpha} \left(-1 + \sqrt{1 - \frac{2\pi^2}{3} \frac{\alpha}{\beta^2}} \right)$$

- We do not understand the reason yet...

Summary

- This means that the I/N corrections may make a difference between the two methods
- There are several possible candidates to resolve this inconsistency